Chapter 3 Carbonyl, Phosphine Complexes

Outline

A. Metal complexes of CO

B. Metal complexes of CO-like ligands

C. Phosphines as ligands

$\left[\ :C\equiv O: \quad :C\equiv N\text{-}R \quad :C\equiv S: \quad :N\equiv O: \quad :N\equiv N:\ \right]$

A. Metal complexes of CO

Why CO chemistry?

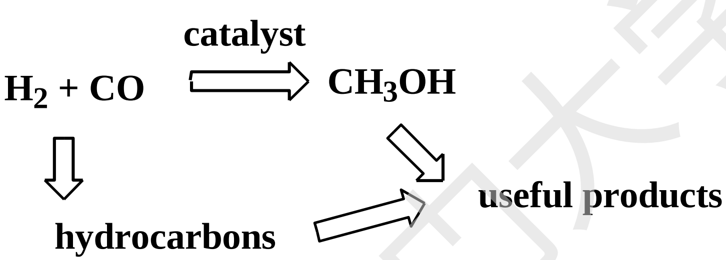
- **Current industrial raw material for carbon-based chemicals:** ***Petroleum oil***
- **Other possible carbon sources:** ***coal, biomass***

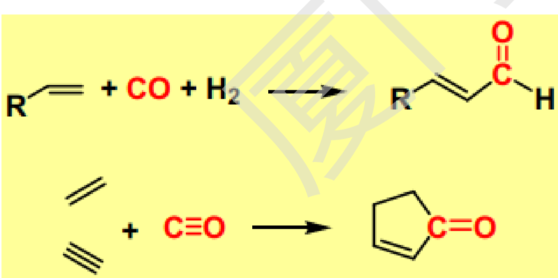
coal, biomass ------------> organic compound ?

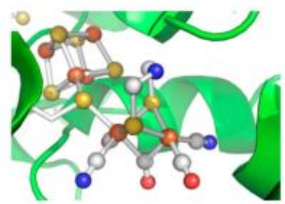
- **A strategy to use coal and biomass for chemical synthesis**

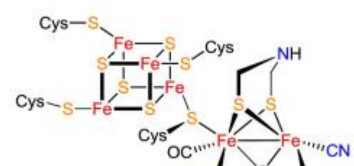
C (coal) + H_2O ---------> H_2 + CO (water gas)

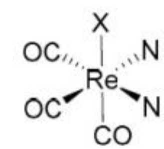
Making compounds from CO

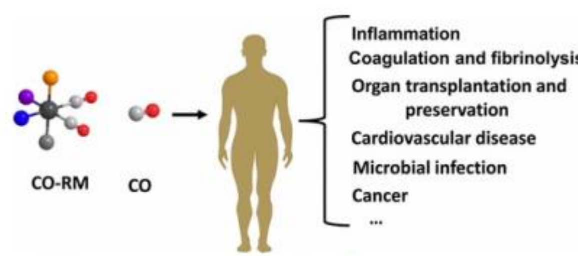




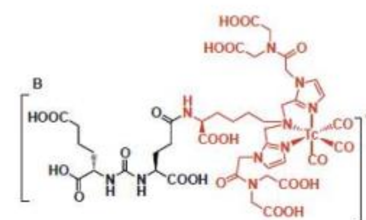



enzyme


catalysis, luminescent

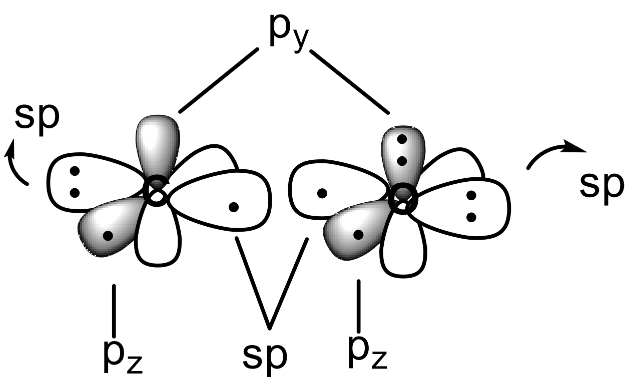


CO-releasing agents for CO therapy


prostate imaging agent

1. Electronic structure of free CO

1) In terms of *valence bond theory*



:C≡O: (with dative arrow ← ·· from O to C) or :C≡O:

This transfer leads to a $\overset{\delta+}{C}–\overset{\delta-}{O}$ polarization of the molecule, which is almost exactly balanced by a partial $\overset{\delta-}{C}–\overset{\delta+}{O}$ polarization of all three bonding orbitals because of the higher electronegativity of oxygen.

Dipole moment:

μ is nearly 0 (0.1 D) , why?

As CO has a nearly 0 dipole moment, it is unreactive.

*** Electronegativity consideration:**

$\overset{\delta+}{:C}\equiv\overset{\delta-}{O:}$

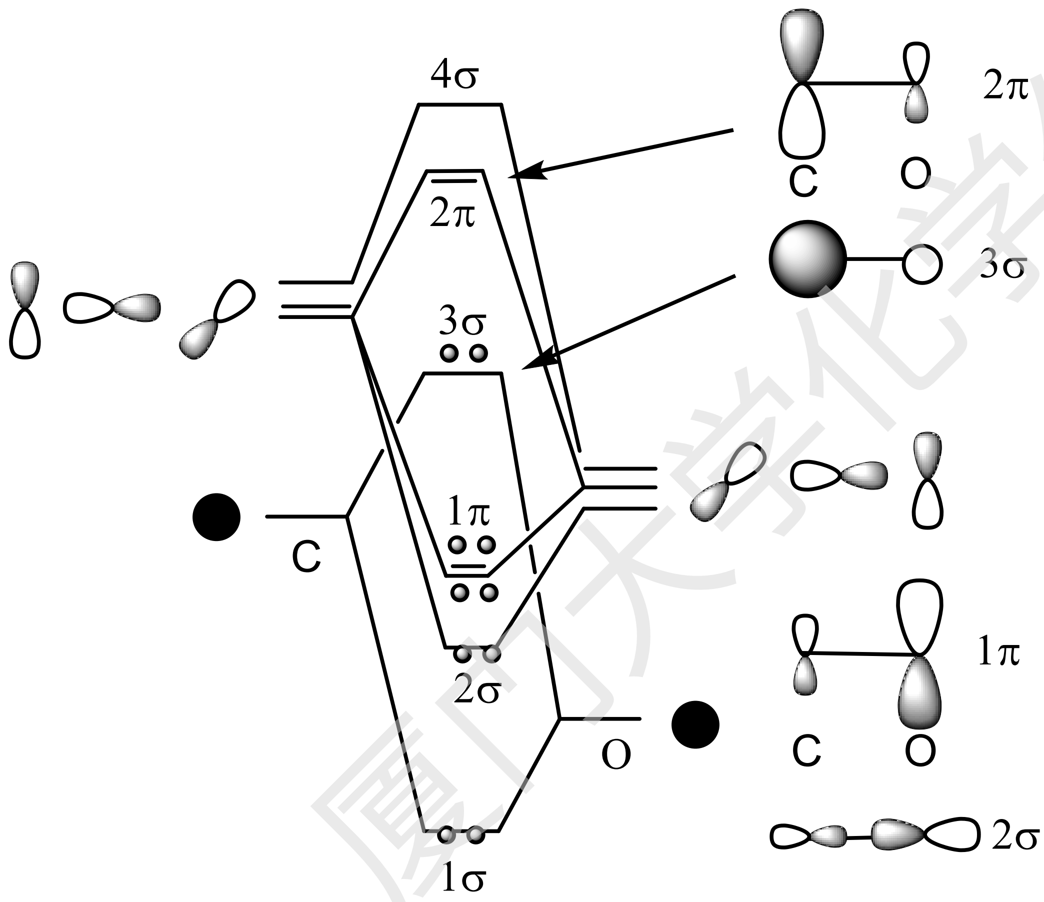
*** Dative bond**

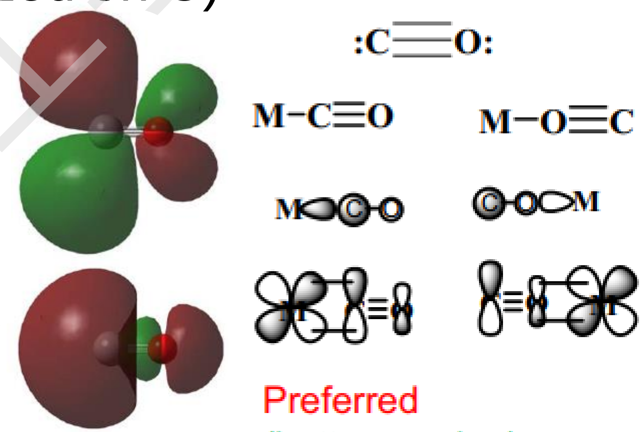
$\overset{\delta-}{:C}\equiv\overset{\delta+}{O:}$

(2) In terms of *MO theory*

Consider s-s, p-p mixing only :

LUMO: 2π orbital (*antibonding character,* mainly localized on C)



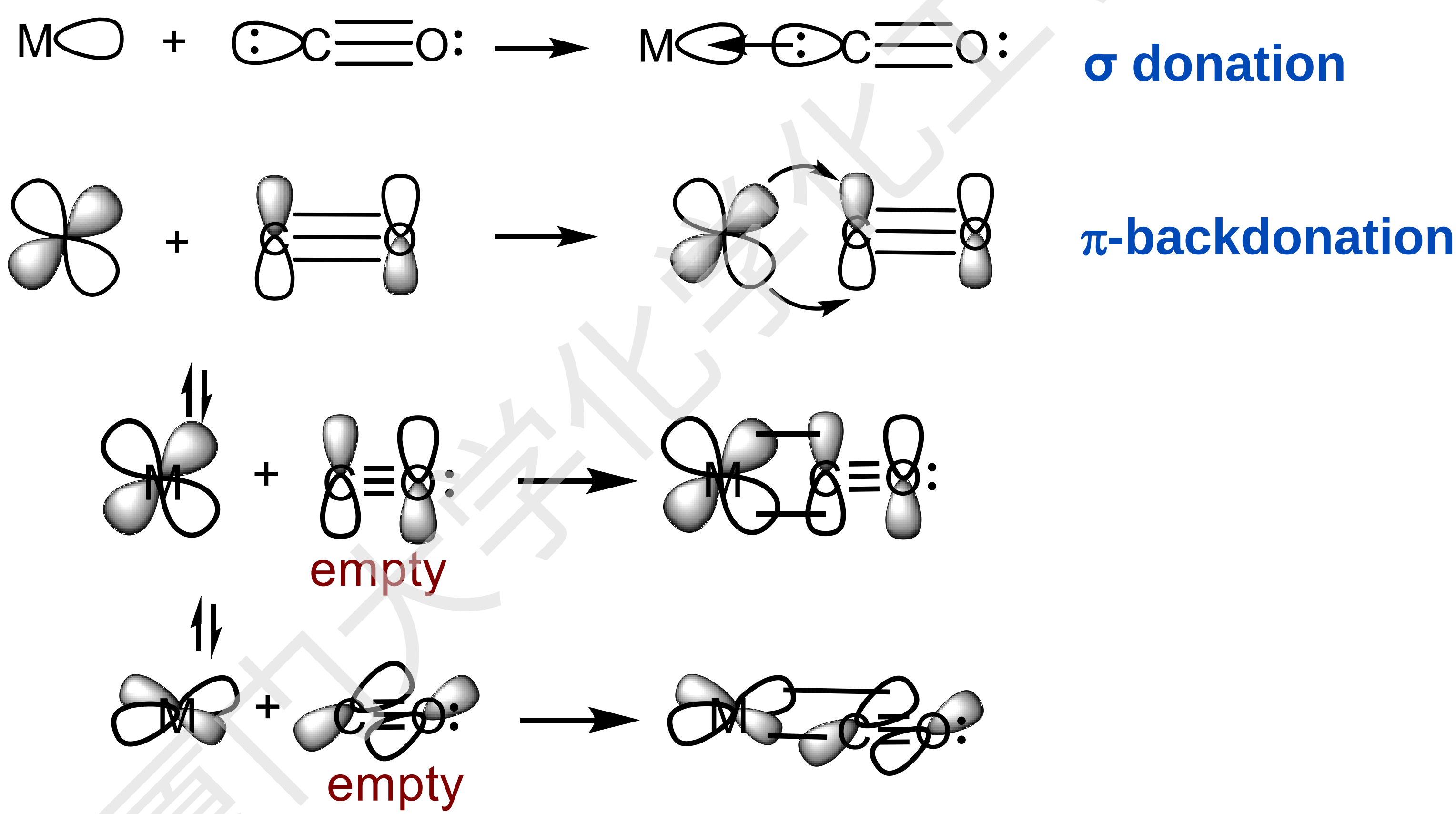


Preferred
(better overlap)

HOMO: 3σ orbital (a little *antibonding character,* mainly localized on C)

2. Electronic structure of CO complexes

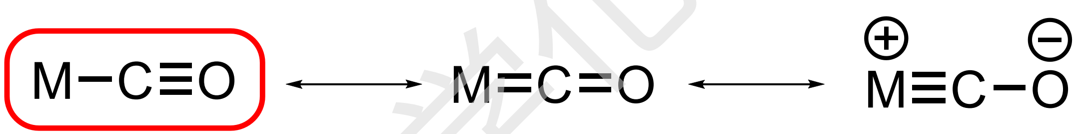
CO is a σ donor, as well as a π acceptor. It can form two π bonds with a transition metal ion.



$\boxed{M{-}C{\equiv}O} \longleftrightarrow M{=}C{=}O \longleftrightarrow \overset{\oplus}{M}{\equiv}C{-}\overset{\ominus}{O}$

π-bonding interaction

- Depending on the metal and other ligands, the nature of the CO ligand in L_nM-CO can vary.
- - If L are good π acceptor or the complex is cationic, then __________ will be more important.
- - If the L are good σ-donors or the complex is anionic, then _______________________will be more important.



(A) (B) (C)

Electron-poor system Electron-rich system

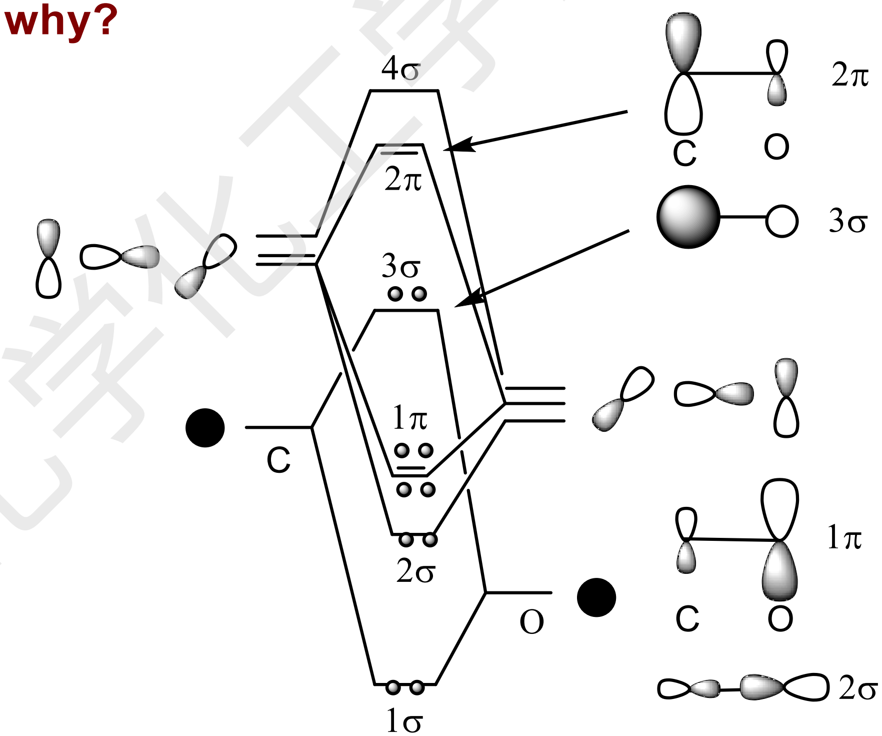
- To form stable CO complexes, π-bonding interaction is often needed.
- Carbonyl complexes of the coinage metals are rare and usually unstable, since the energy of d electron is too low for π-back bonding
- Positive charged carbonyl complexes usually have relatively labile CO, since the energy of d electron is too low for π-back bonding

Why C？

In metal complexes, we usually observed

M—C≡O not M—O≡C **why?**

:C≡O:

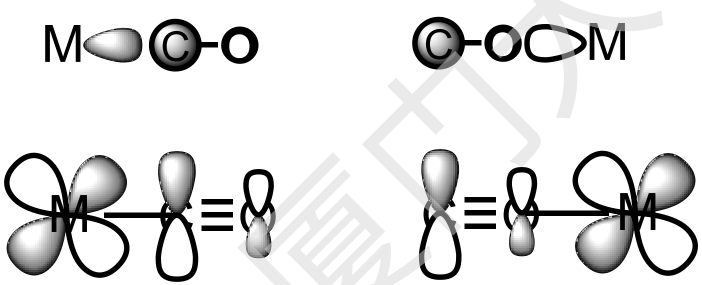


Electronegativity consideration:

C is less electronegative

In term of orbital interaction:

M−C≡O C≡O−M

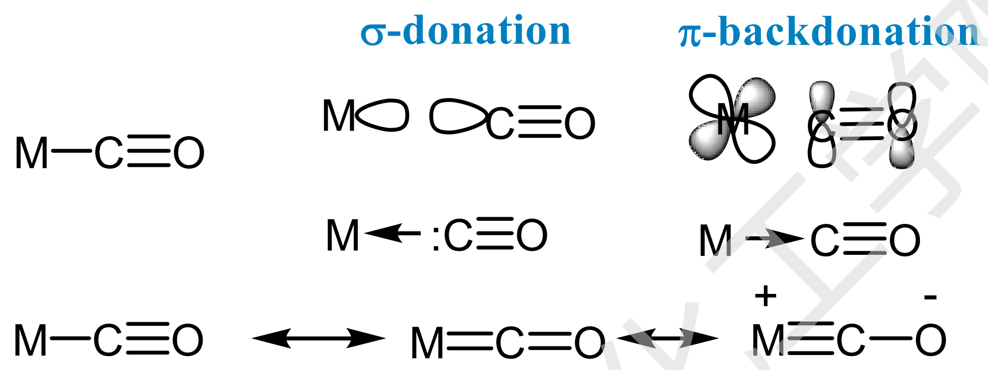


better overlap

The HOMO is 3σ. The lone pair is more or less localized on carbon.

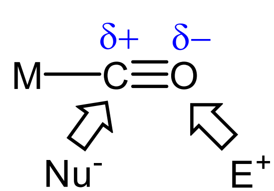
The LUMO 2π can interact with a d orbital of M better if C is attached to metal.

Effect of complexion



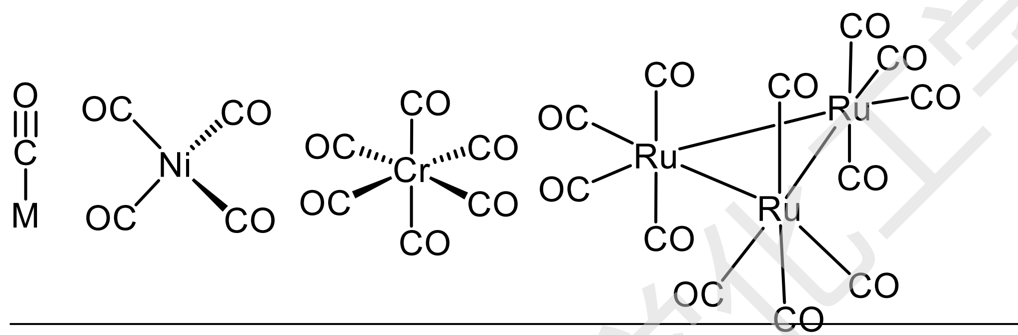
	σ donation	π backdonation	overall
d(M-C)	↓	↓	↓
d(C-O)	↓	↑	↑
υ(CO)	↑	↓	↓
e- density on C	↓	↑	↓
e- density on O	↓	↑	↑

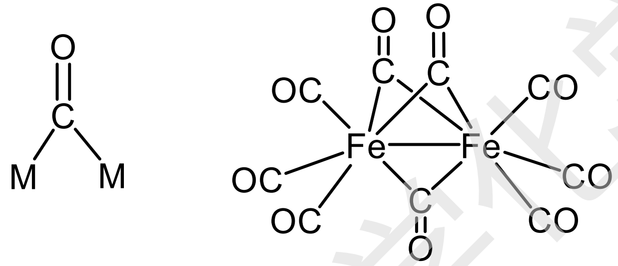
Expected reactivity:



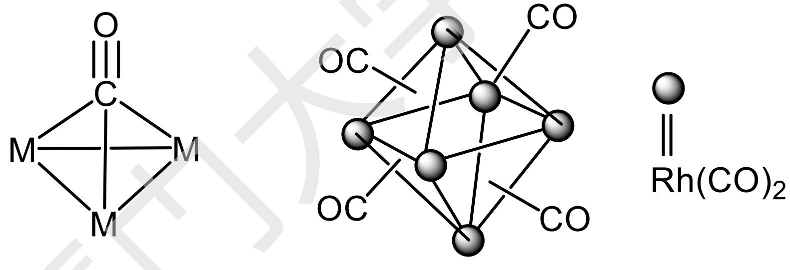
3. Structures of M-CO compounds

***The three principal coordination modes of CO*:**





μ_2-complexes



$Rh_6(CO)_{16}$

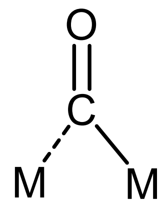
μ_3-complexes

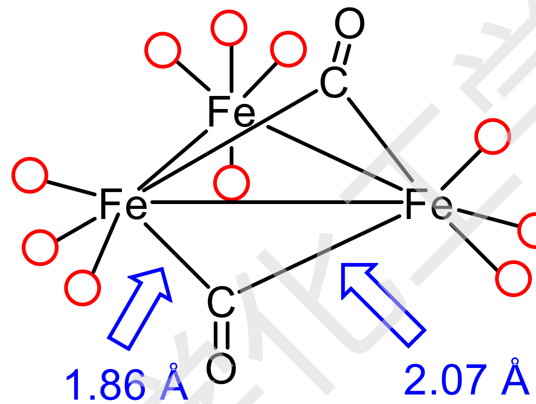
similar to M-H-M

Other less common coordination modes

Semibriding CO

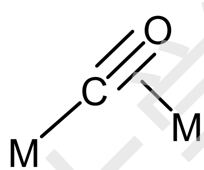
Just for information

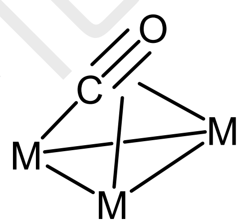


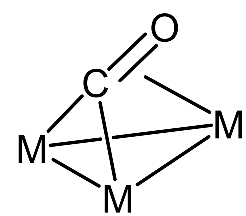


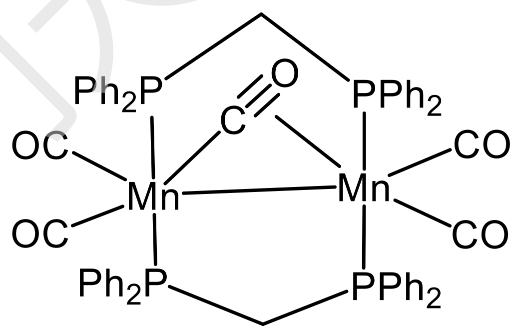
$Fe_3(CO)_{12}$

σ/π- bridge CO

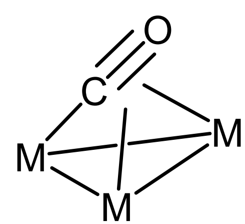








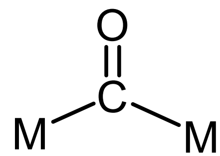
$Mn_2(dppm)_2(CO)_5$

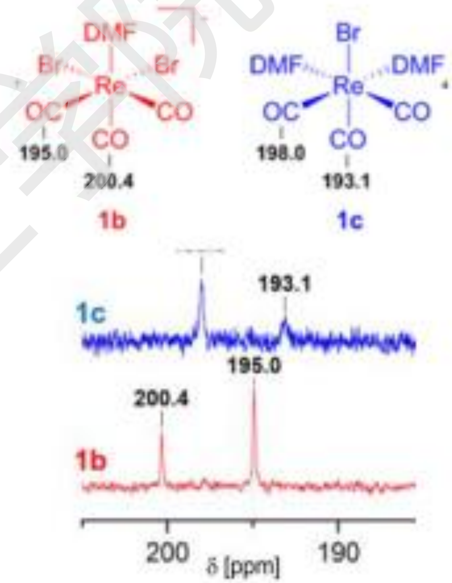


M = $CpNb(CO)_2$

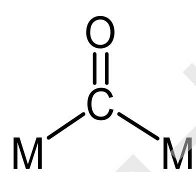
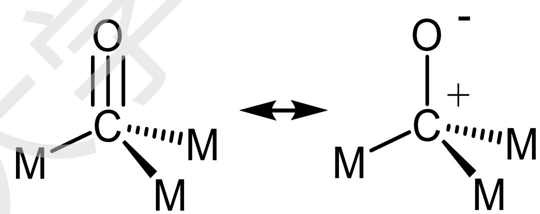
4. Characterization of M-CO compounds

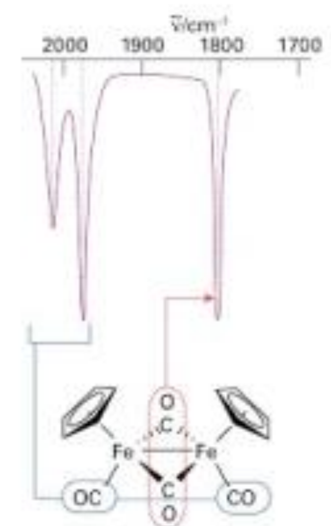
^{13}C NMR

	M—C≡O	
δ(CO), ppm:	150-220	230-280



IR: υ(CO), cm^{-1}

C≡O	M—C≡O		
2143	1850-2150	1700-1860	1600-1700

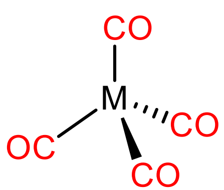


IR data of selected carbonyl complexes, υ(CO), cm^{-1}

$V(CO)_6$ 1976 cm^{-1}	$Cr(CO)_6$ 2000 cm^{-1}	$Mn_2(CO)_{10}$ 2013 cm^{-1}	$Fe(CO)_5$ 2023 cm^{-1}	$Co_2(CO)_8$ 2044 cm^{-1}	$Ni(CO)_4$ 2057 cm^{-1}
$[Ti(CO)_6]^{2-}$ 1747 cm^{-1}	$[V(CO)_6]^{-}$ 1860 cm^{-1}	$[Cr(CO)_6]$ 2000 cm^{-1}	$[Mn(CO)_6]^{+}$ 2090 cm^{-1}	$[Fe(CO)_6]^{2+}$ 2204 cm^{-1}	CO 2143 cm^{-1}

Back - bonding

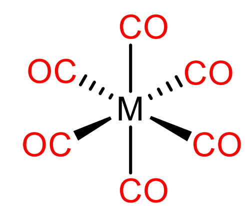
decreases



$Ni(CO)_4$	2060
$Co(CO)_4^-$	1890
$Fe(CO)_4^{2-}$	1790

increases

$Mn(CO)_6^+$	2090
$CrCO)_6$	2000
$V(CO)_6^-$	1860

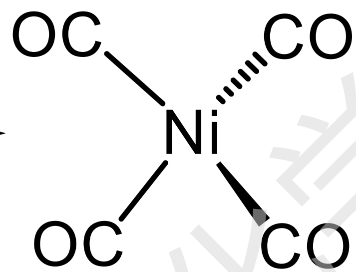


5. Preparation of CO complexes

Can be made from CO and O-containing organic compounds

1) From CO

- **Metal + CO**

Ni + 4 CO $\longrightarrow$  18e

Colorless liquid, b. p. = 34 ºC

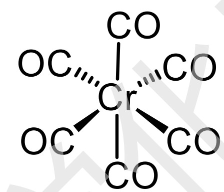
$Fe + 5\ CO \longrightarrow Fe(CO)_5$ 18e

straw-colored liquid, b. p. = 103 ºC

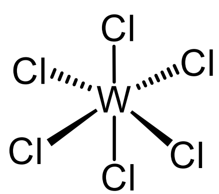
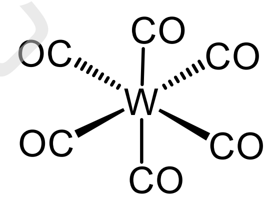
5. Preparation of CO complexes

- **Metal Salts + reducing agent + CO ==> 18e complexes**

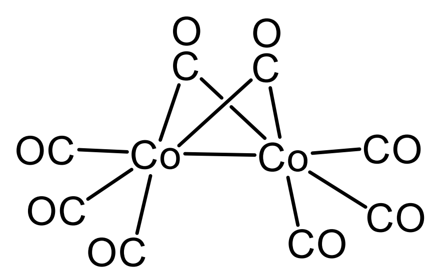
Typical reducing agents: Na, Al, H_2, AlR_3, CO ...

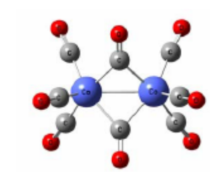
$CrCl_3 + Al + 6\ CO \longrightarrow$  18e

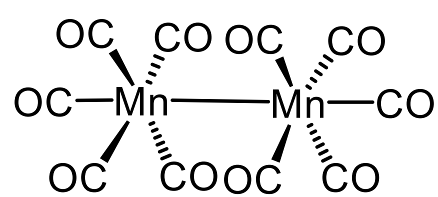
$VCl_3 + Na + CO \longrightarrow NaV(CO)_6$ 18e

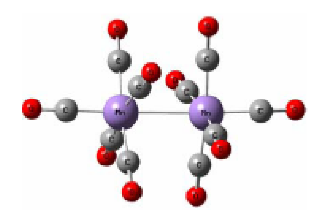
 $+ AlEt_3 + CO \longrightarrow$  18e

How?

$CoCO_3 + H_2 + CO \longrightarrow$ 

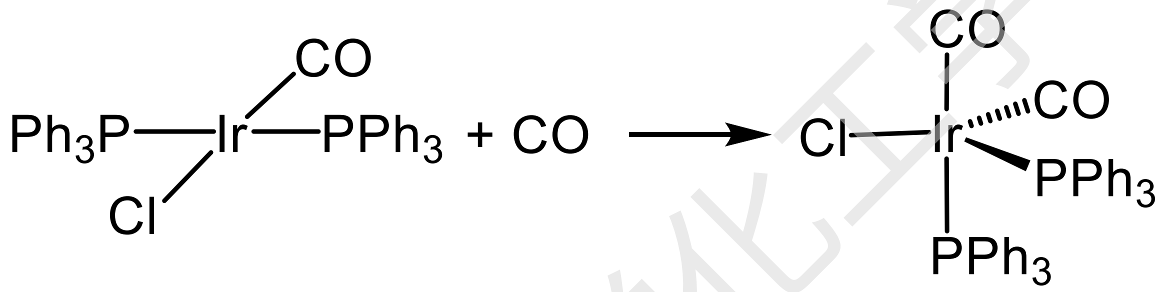


$Mn(OAc)_2 + CO + AlEt_3 \longrightarrow$ 



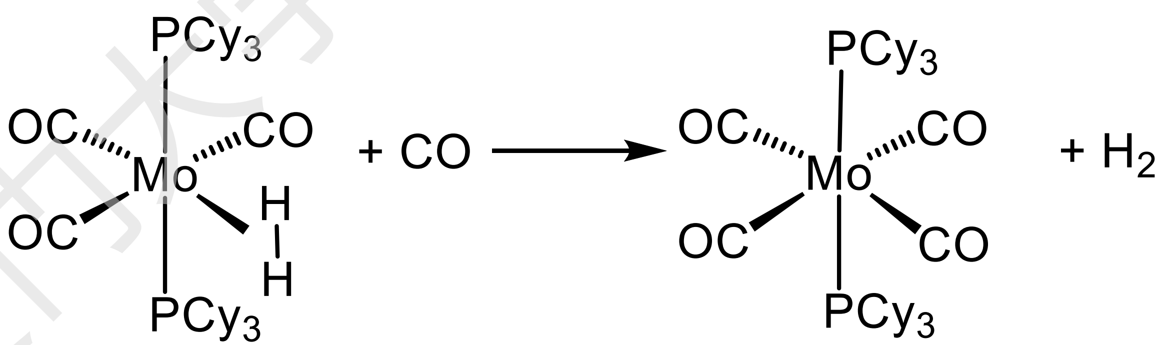
5. Preparation of CO complexes

- **Unsaturated compounds + CO**



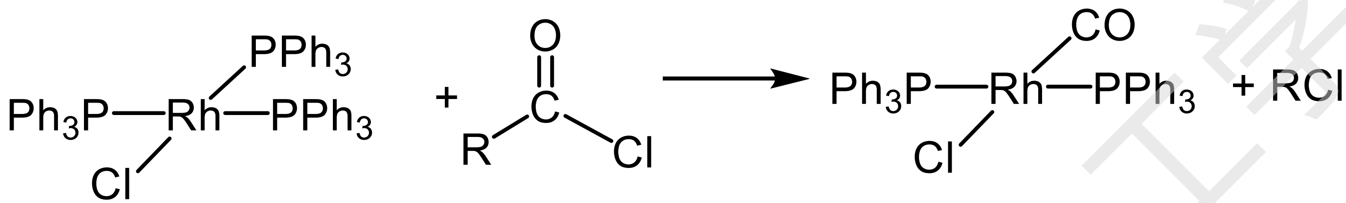
16e 18e

- **Substitution reactions**



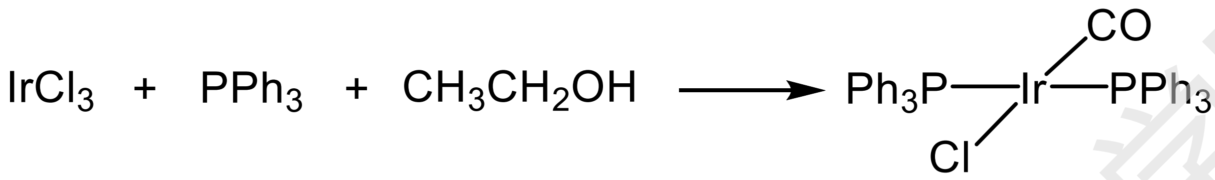
5. Preparation of CO complexes

2) From a reactive organic carbonyl compound.



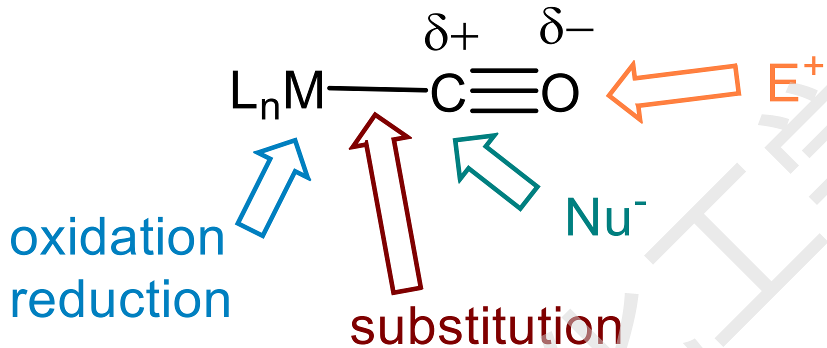
Mechanism?

Example

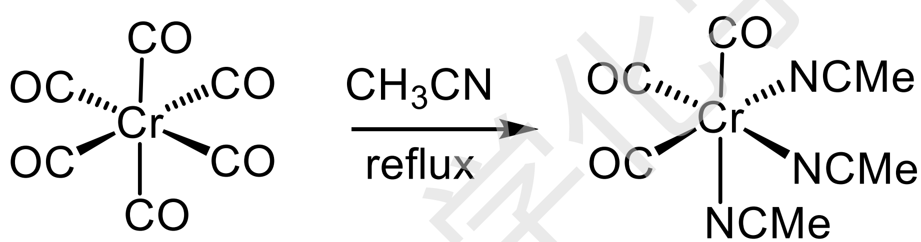


Mechanism?

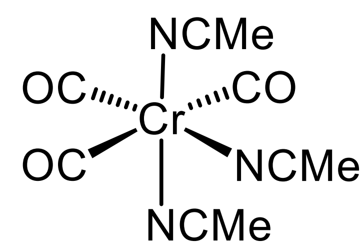
6. Reactions of metal carbonyls



(1) Substitution

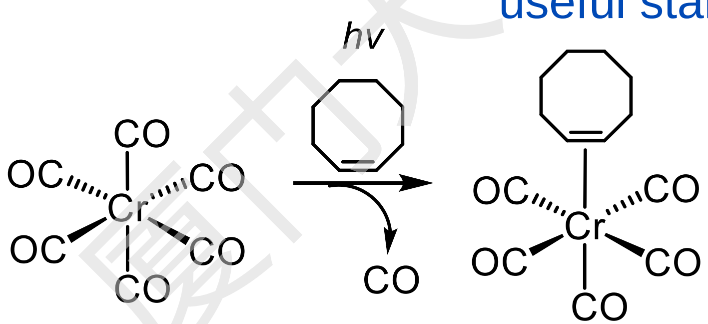


useful starting material



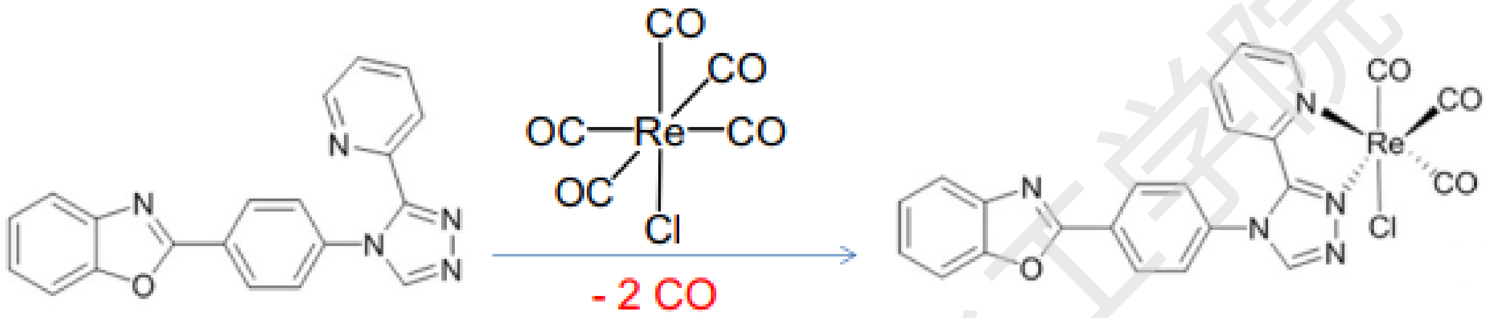
not

Why?

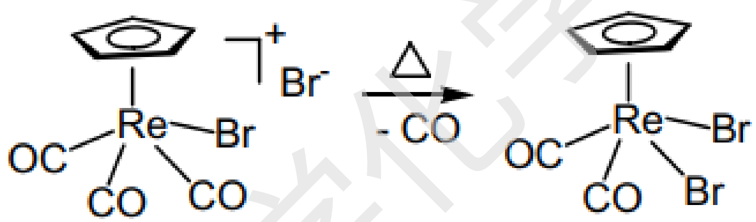


- CO ligands in neutral complexes are usually tightly bounded.
- Harsh condition is need for the substitution reactions.

Examples



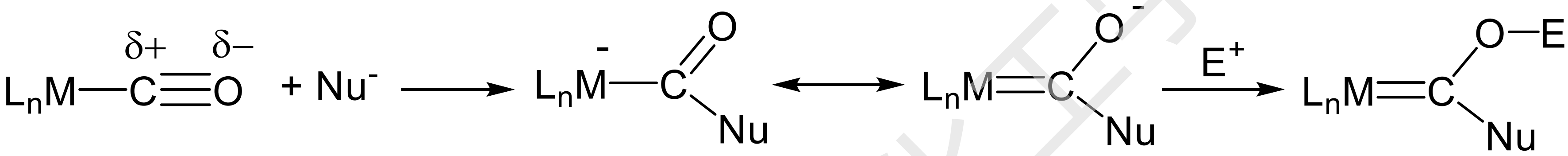
Chelating effect facilitated the substitution reaction

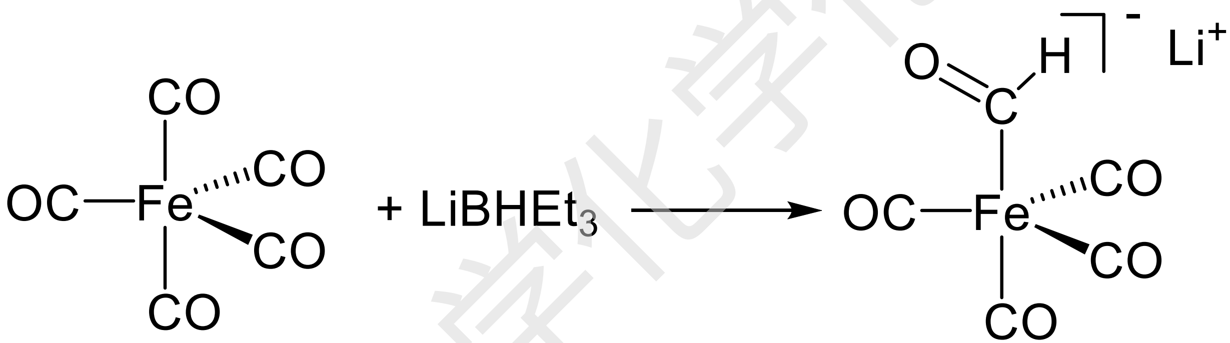


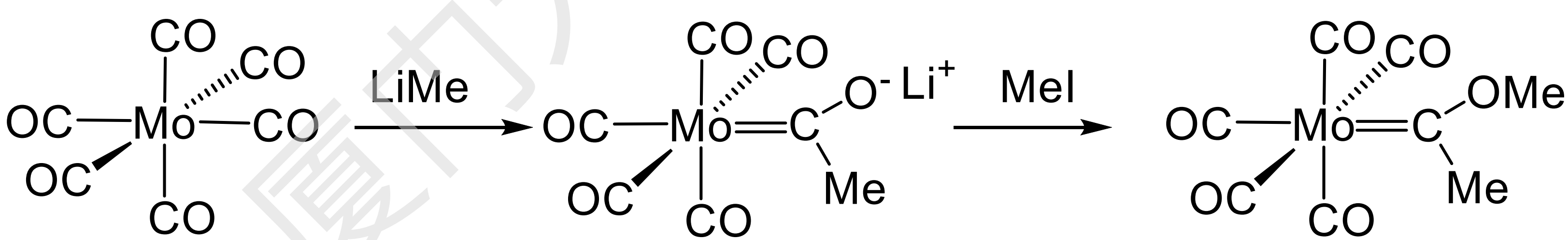
CO-substitution reactions of cationic complexes could be easier than those of neutral ones

There are many factors determining the lability of CO in CO complexes, e.g. π-backbonding, d configuration, other ligands present.

(2) Nucleophilic attack on carbon

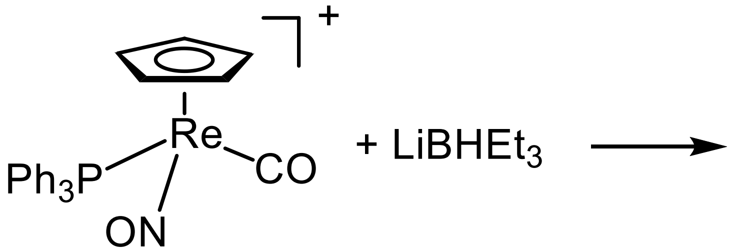


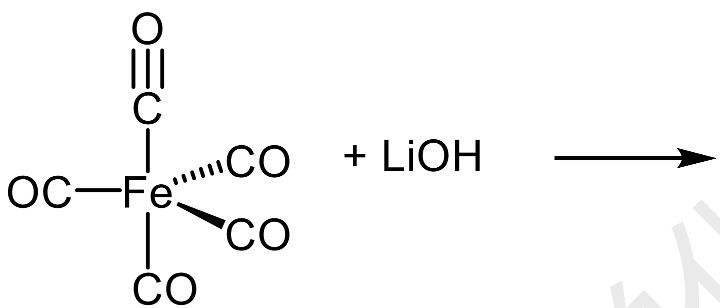


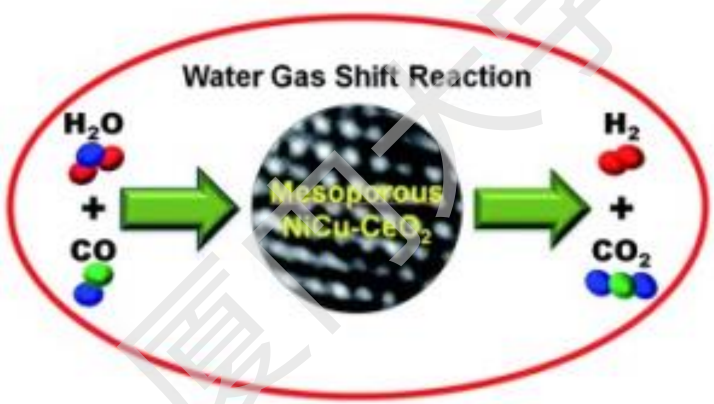


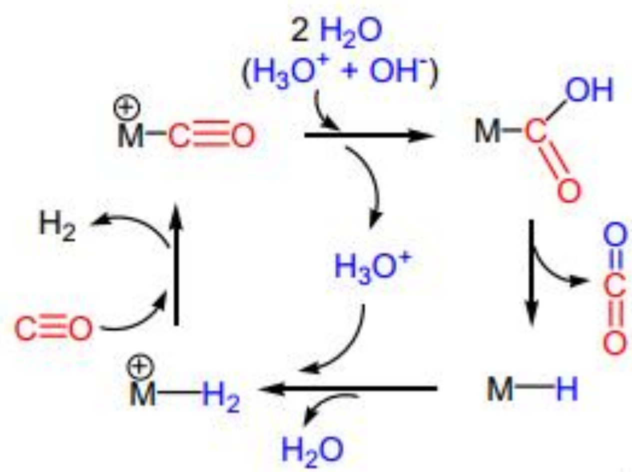
Practice Problems

Provide the products for the following reactions.





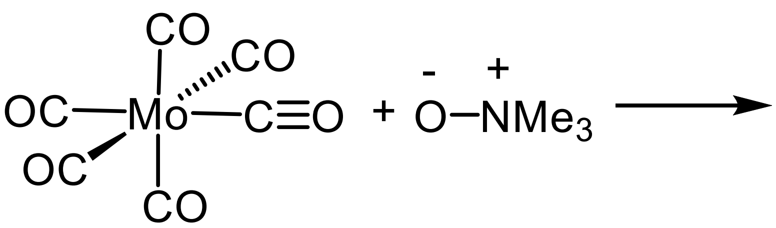




Note: The reaction can be used to explain the mechanism of water-gas shift reaction

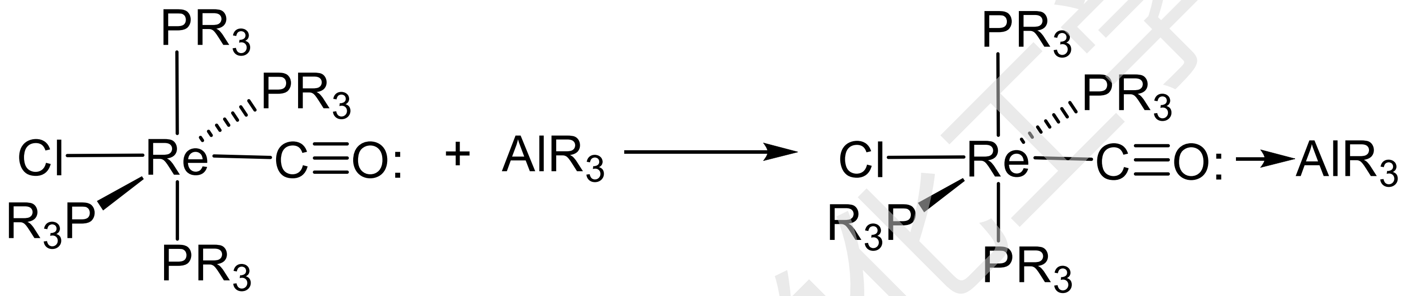
Practice Problems

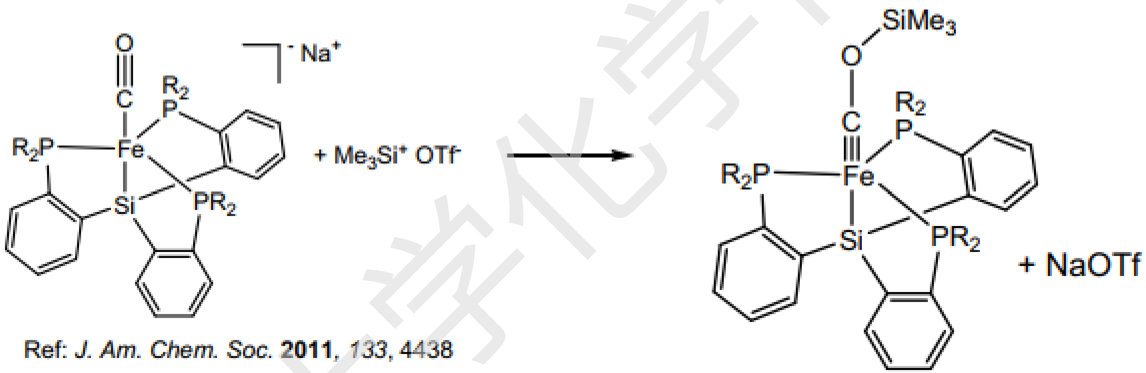
Provide the products for the following reactions.



A very useful reaction that can be used to remove CO

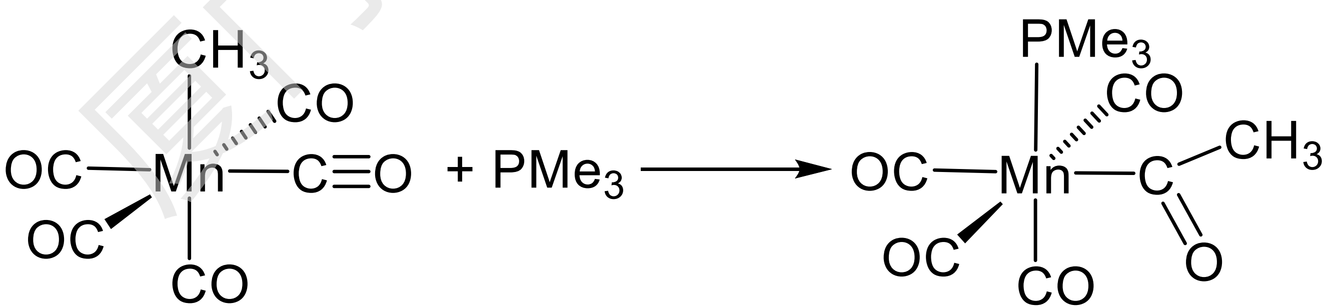
(3) Electrophilic attack at oxygen



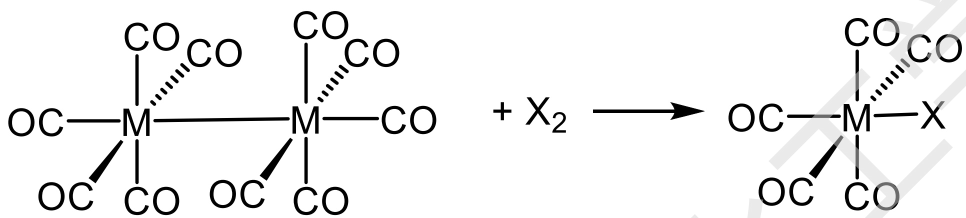


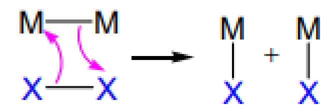
Ref: *J. Am. Chem. Soc.* **2011**, *133*, 4438

(4) Migratory insertion reactions



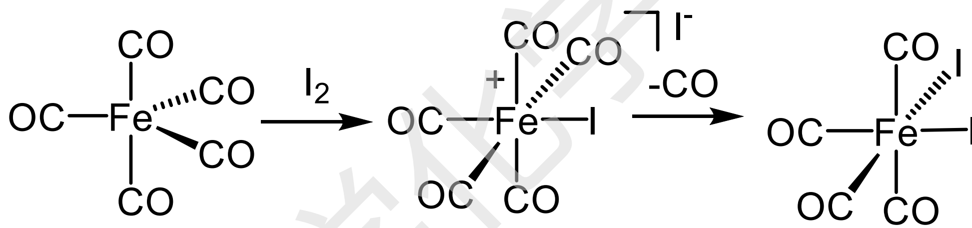
(5) Oxidation



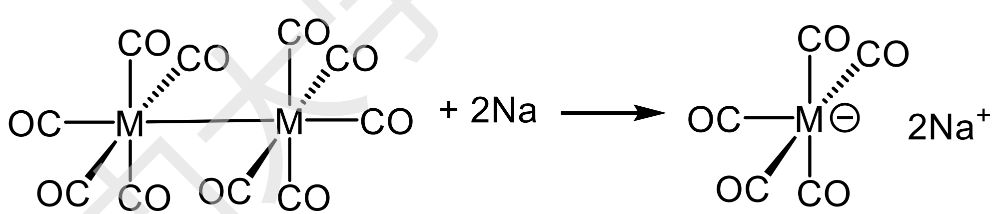


M = Mn, Tc, Re   X = H, Cl, Br, I

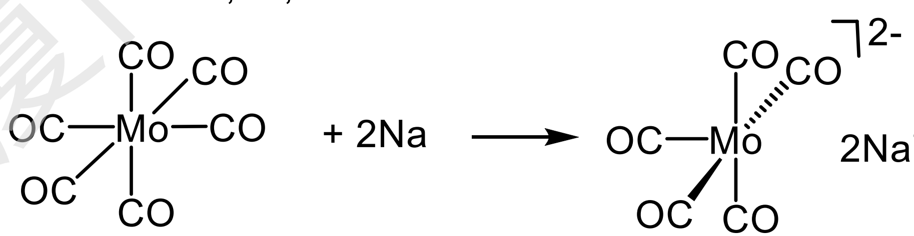
M(0)   M(I)



(6) Reduction



M = Mn, Tc, Re



B. Metal complexes of CO-like ligands

$$\left[\ :C\equiv O: \quad :C\equiv N\text{-}R \quad :C\equiv S: \quad :N\equiv O: \quad :N\equiv N:\ \right]$$

1. Isocyanide complexes (metal isonitrile)

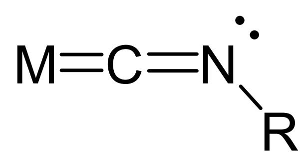
$$M-C\equiv O \qquad M-C\equiv N-R$$

- In terms of composition and structures, metal isonitriles are very similar to metal carbonyls. *e.g.*

$M(CO)_6$	$M(CO)_5$	$M_2(CO)_{10}$	$M(CO)_4$
$M(CNR)_6$	$M(CNR)_5$	$M_2(CNR)_{10}$	$M(CNR)_4$
M = Cr, Mo W	Fe, Ru, Os	Mn, Tc, Re	Ni, Pd, Pt

$$M-C\equiv N-R$$

4.8

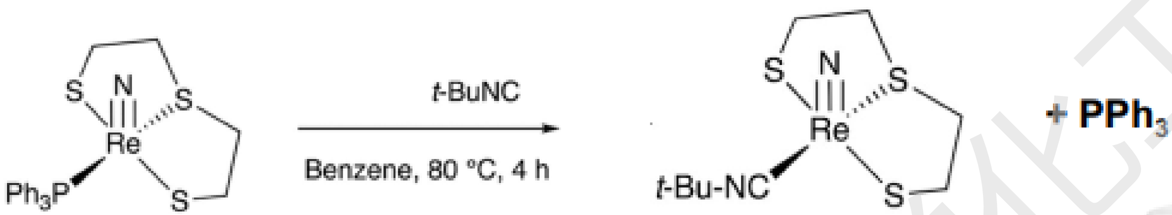


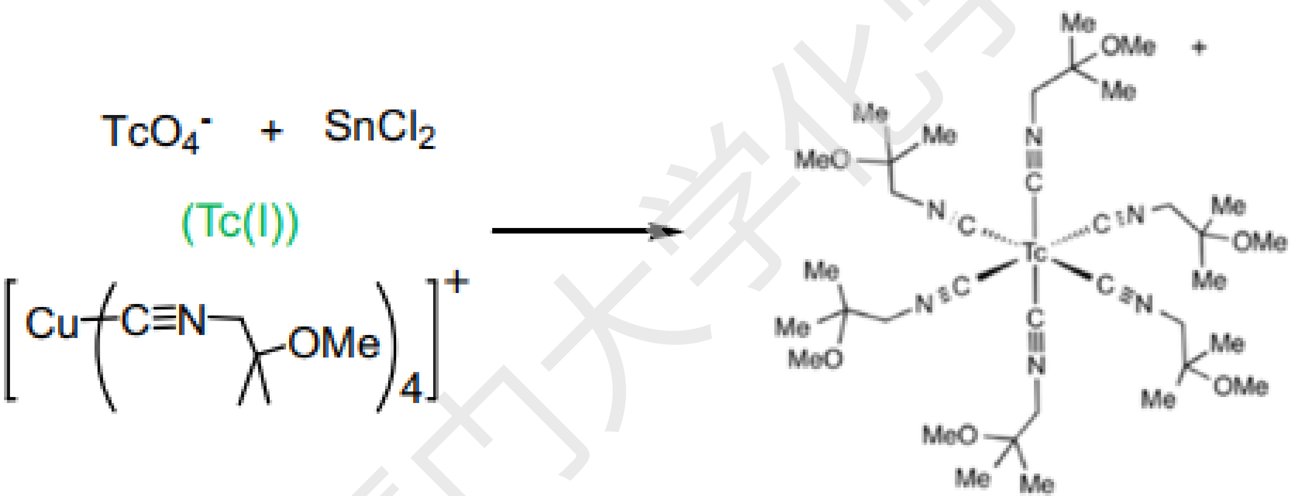
4.9

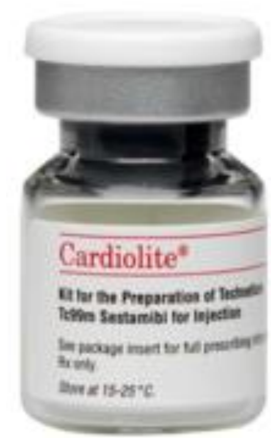
1. Isocyanide complexes (metal isonitrile)

- **Synthesis of isocyanide complexes**

They are usually obtained by ligand substitution reactions of RN≡C.



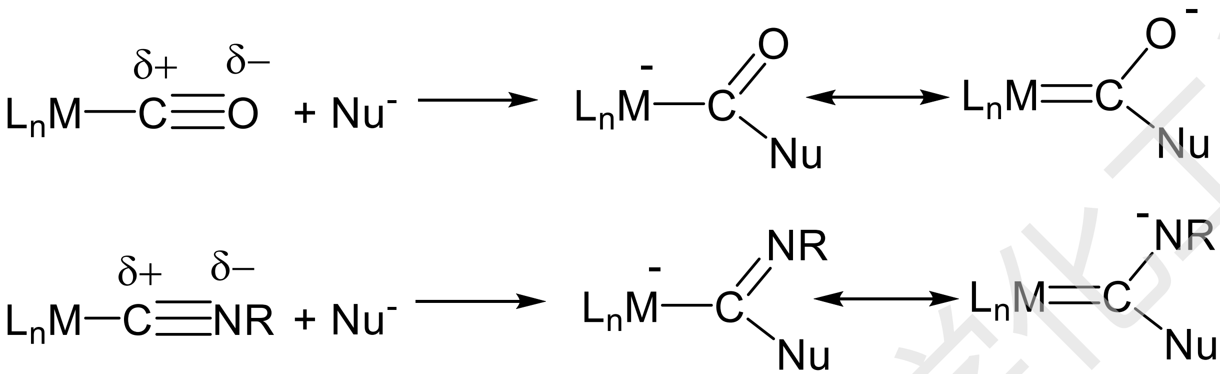




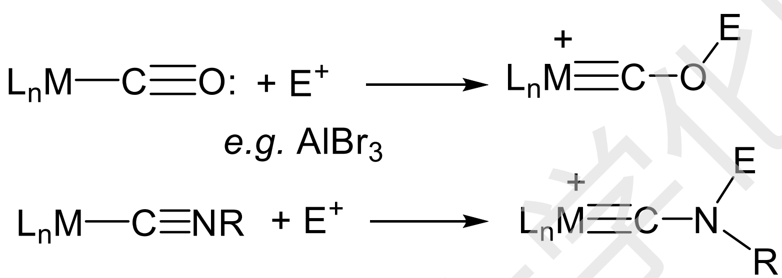
Cardiolite® (technetium Tc-99m sestamibi), an extremely successful myocardial imaging agent, has been used routinely for more than thirty years.

1. Isocyanide complexes (metal isonitrile)

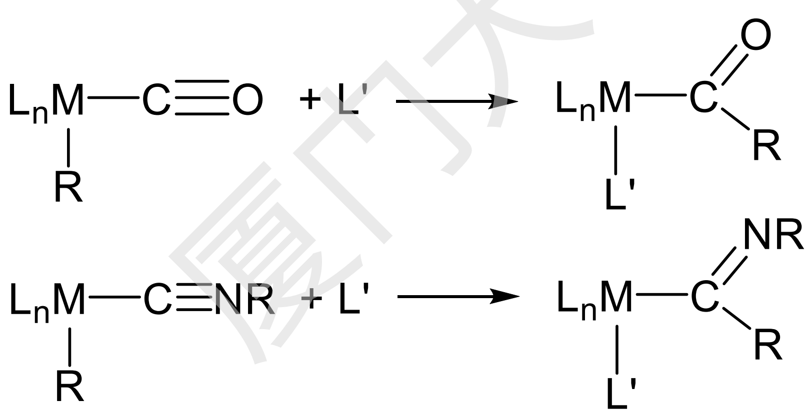
- **In terms of chemical properties, they are also very similar.**



Nucleophilic attack on carbon



Electrophilic attack at oxygen

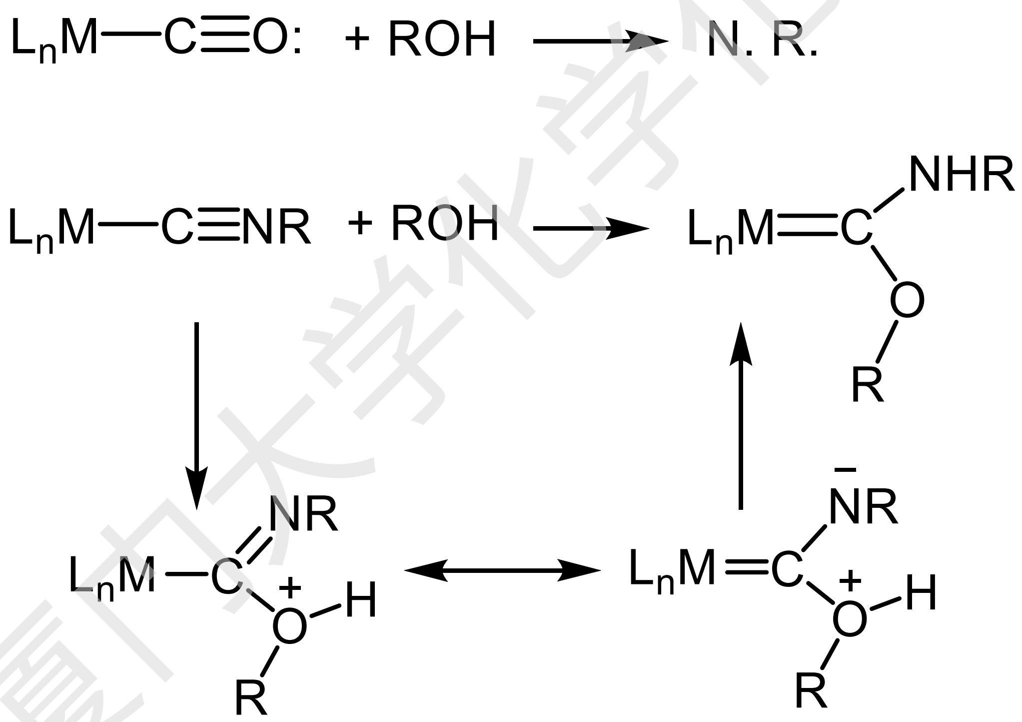


Migratory insertion reactions

1. Isocyanide complexes (metal isonitrile)

- **Difference between complexes of CO and CN-R.**

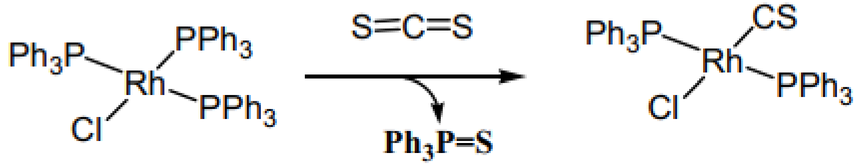
M-CNR complexes are more reactive toward both nucleophiles and electrophiles.



2. CS complexes

C≡S unstable (more reactive), but M-C≡S can be stable

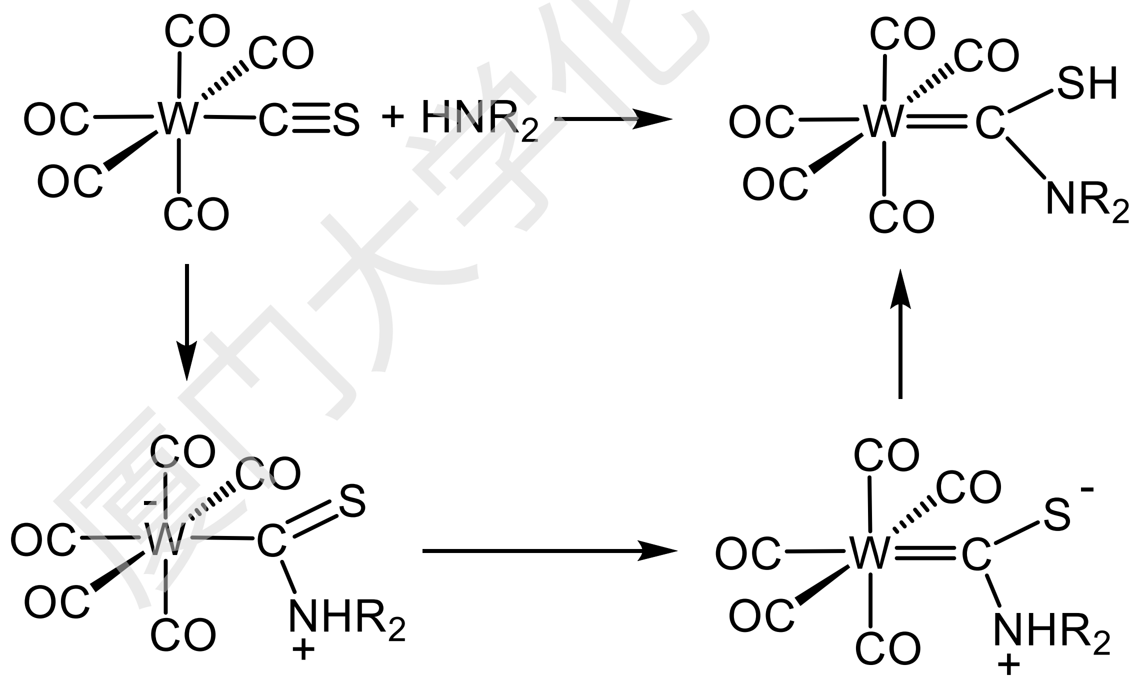
- CS complexes could often be obtained by de-sulfurization of CS_2



- Chemical properties of M-CS complexes:

✓ Similar to those of CO and M-CNR complexes

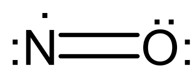
✓ More reactive than CO complexes



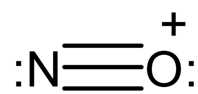
3. NO (nitrosyl) complexes

- ***Lewis structure of NO***

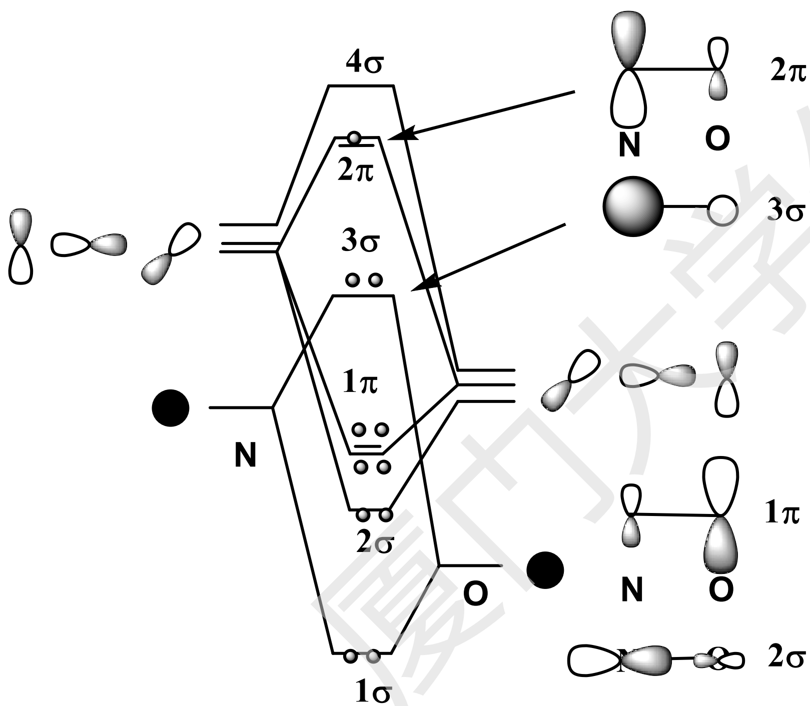
NO



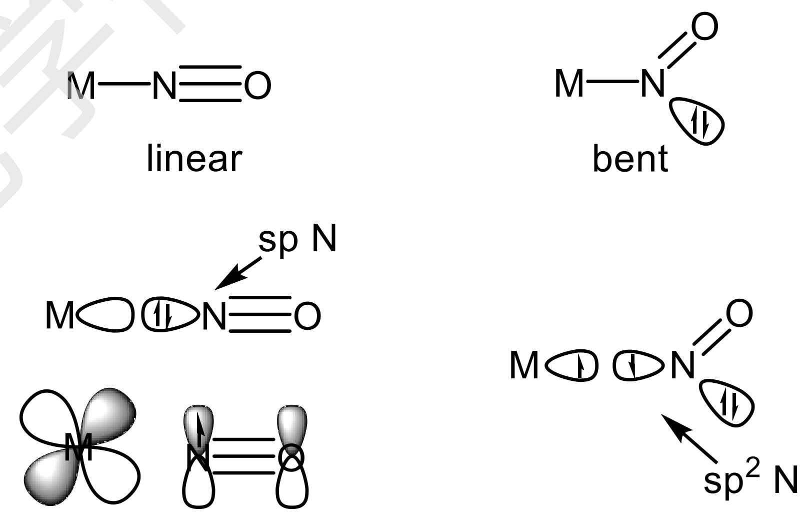
NO$^+$ (isoelectronic to CO)



- ***MO description of NO***

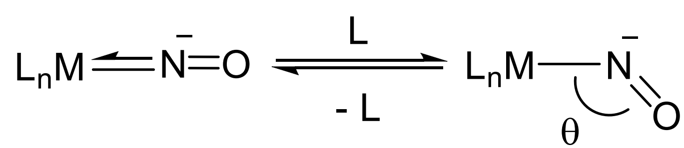


- ***Bonding in M-NO.***



3e donor

1 e donor

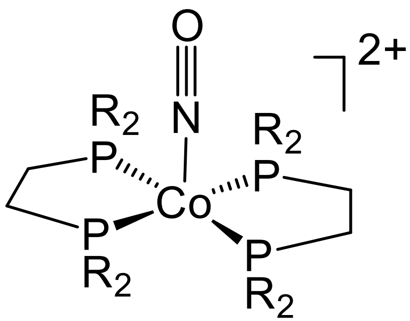


Linear NO: θ ≈ 160-180°

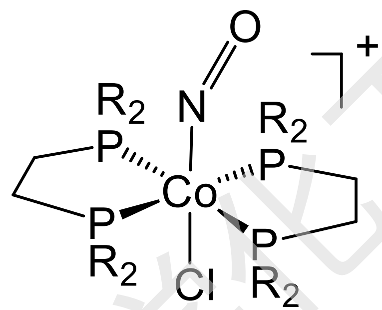
Bent NO: θ ≈ 120-140°

Practice Problems

1. What is the electron count for the following complexes?



A



B

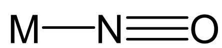
Co	**9**
4-PR$_2$	**8**
NO	**3**
2+	**-2**
	-18e

Co	**9**
4-PR$_2$	**8**
NO	**1**
Cl	**1**
1+	**-1**
	-18e

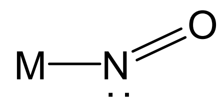
Oxidation state of Co?

Practice Problems

2. How could we differentiate linear or bent NO ?



linear



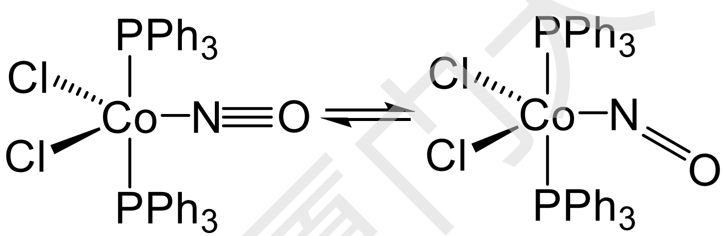
bent

IR, υ(NO), cm^{-1} **1610-1830** **1520-1720**

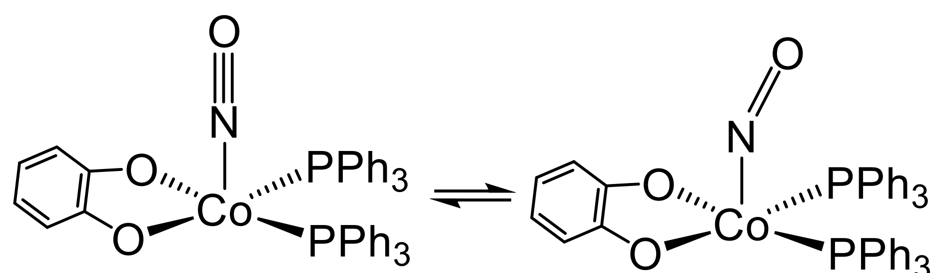
Free NO	ν_{NO} 1876 cm^{-1}
Free NO^+	ν_{NO} 2250 cm^{-1}
NO ligand	ν_{NO} 1900 - 1600 cm^{-1}

3. Can we predict if a NO is linear to bent?

The two isomers can have very similar energy. e.g.



18e 16e



18e 16e

Preparation of NO complexes

from NO gas and NO+ (e.g. NOBF4)

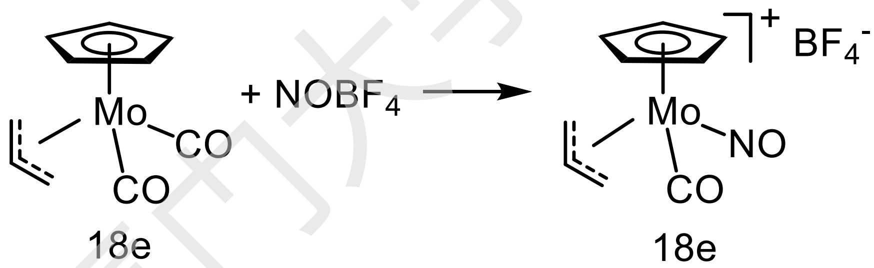
(1) From NO gas. e.g.

$Cr(CO)_6$ + excess NO -----> $Cr(lin\text{-}NO)_4$ + 6 CO

$Fe(CO)_5$ + excess NO -----> $Fe(CO)_2(NO)_2$ + 3 CO

Note: 2 NO for 3 CO. **Why?**

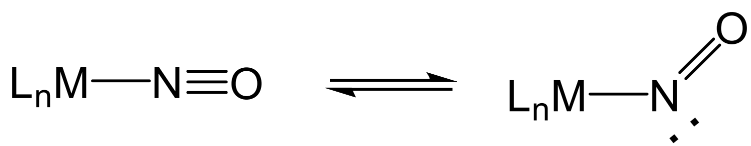
(2) From $NOBF_4$, *e.g.*



Note: 1 NO^+ for 1 CO.

Some chemical properties of M-NO complexes

a. Line M-NO and bent M-NO interconversion

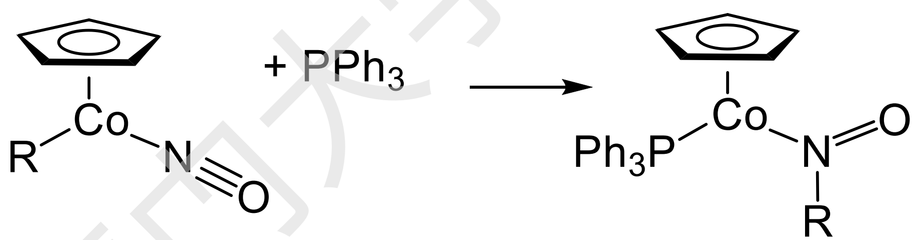


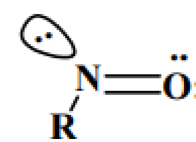
18e 16e

important in substitution reactions.

b. They are less reactive towards nucleophiles and electrophile than CO.

c. M-NO complexes can also undergo insertion reactions.

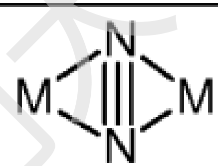
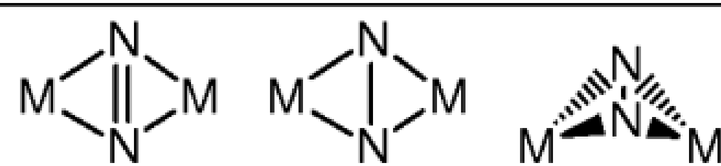
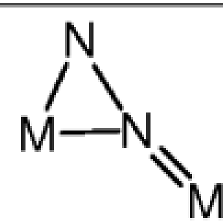




VE = ?

4. CN- and N_2 (dinitrogen) complexes

- Many N_2 complexes are known, and have been extensively studied for N_2- fixation.
- N_2 can be a terminal or a bridging ligand.
- Terminal N_2 complexes have chemical properties similar to those of CO, but are much less reactive.

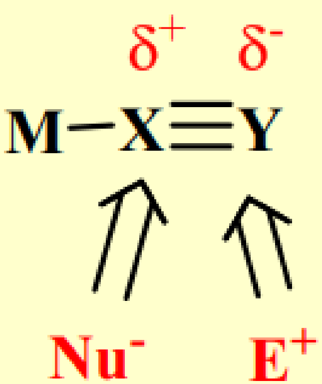
N_2 Binding Mode	Weak Activation	Strong Activation
End-on mononuclear	$M-N\equiv N$	
End-on dinuclear	$M-N\equiv N-M$	$M=N-N=M$
Side-on dinuclear		
Side-on/End-on dinuclear		

Summary

CO-like complexes

Chemical property:

Similar to CO complexes



$M-C\equiv S$

$M-C\equiv N-R$

$M-C\equiv O$

$M-N\equiv O$

$M-N\equiv N$

relative reactivity

less reactive

C. Phosphines as ligands

Phosphines are probably the most important organometallic ligands because their steric and electronic properties can be easily modified.

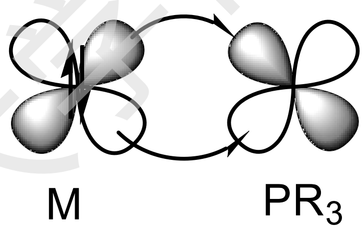
1. Nature of M-PR_3 bonds

PR_3 can function as σ-donor

$$M \longleftarrow :PR_3 \quad \Rightarrow \quad M—PR_3$$

It can function as a π-acceptor. **How?**

Early belief.



However:

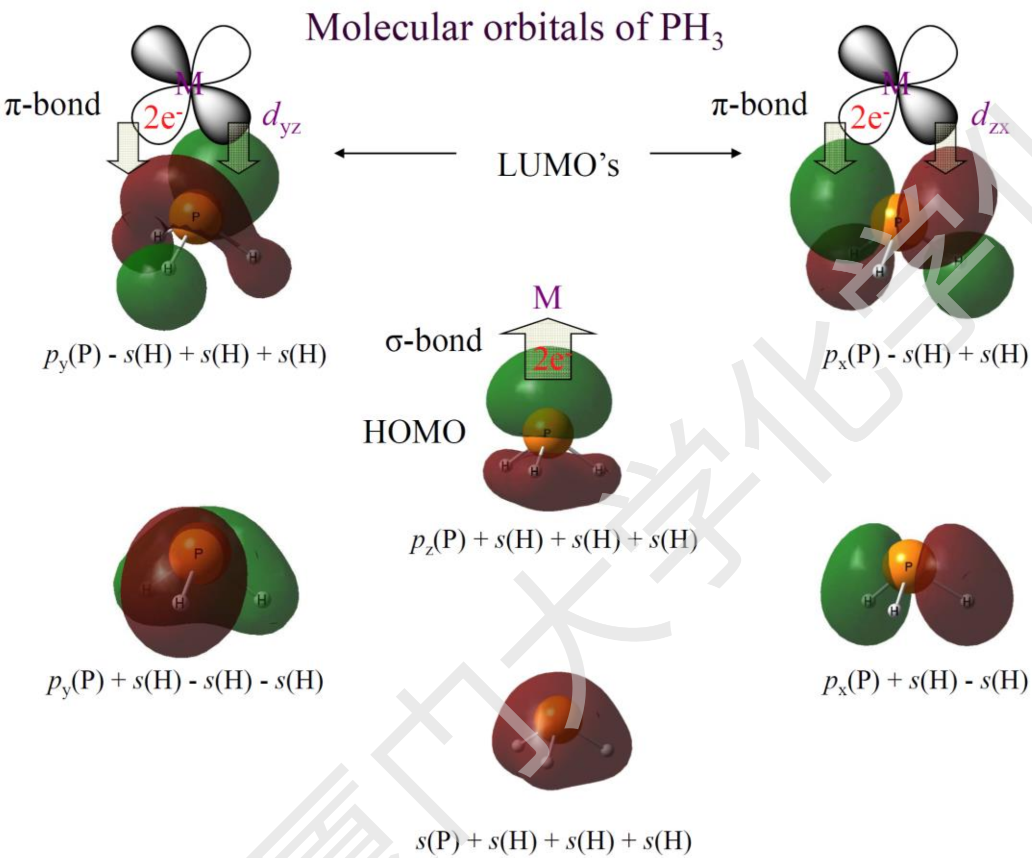
* No experimental evidence supports it.

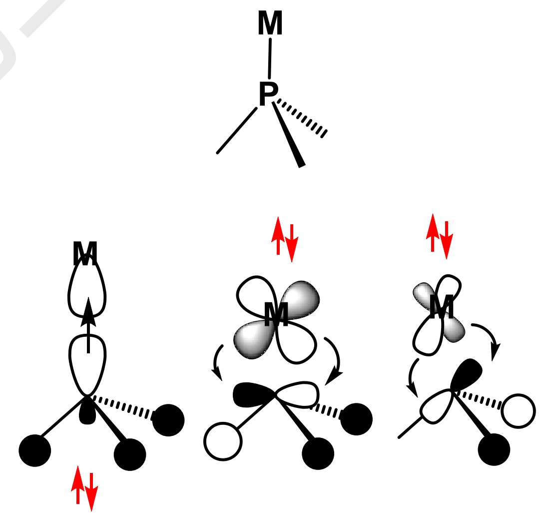
* Theoretical calculation shows that dπ-pπ interaction is not important.

1. Nature of M-PR_3 bonds

Current belief:

The π-accepting properties of PR_3 is due to d(M) -> σ^*(P-R).

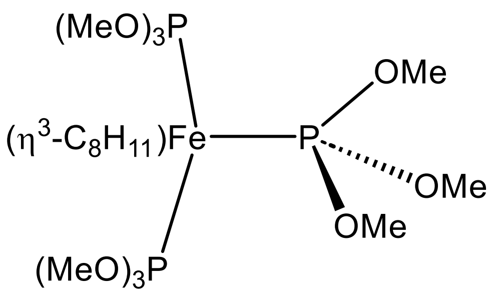




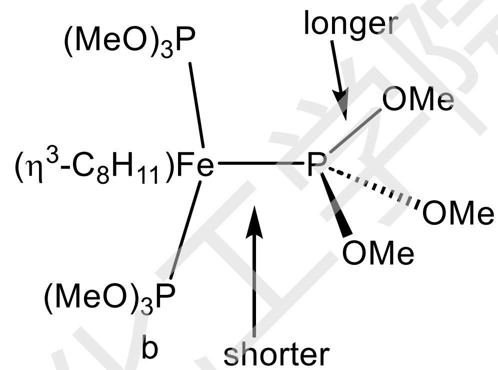
Phosphines, like amines, are strong σ-donors.
Unlike amines, they are also π-acceptors.

Evidences for dπ (M)->σ* (P-R) bonding

Evidence 1.



a



b

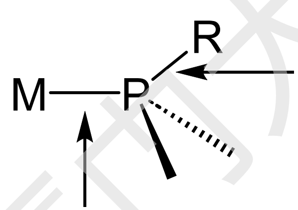
$d(P\text{-}O)_a - d(P\text{-}O)_b = -0.021$ Å

$d(M\text{-}P)_a - d(M\text{-}P)_b = 0.015$ Å

As a result, the backdonation from metal to PR_3 plays an important role.

If there is dπ(M)->σ*(P-R) bonding,

Why?



P-R bond becomes longer or shorter?

M-P bond becomes longer or shorter?

2. Electronic property of PR_3

Q1. Consider the ligands PCl_3, PPh_3, $PMePh_2$, PMe_3.

What is the order of σ-donating ability ?

What is the order of π-accepting ability?

- σ-donating ability: **PCl_3 < PPh_3 < $PMePh_2$ < PMe_3**
- π accepting ability: **PCl_3 >PPh_3 > $PMePh_2$ > PMe_3**

In general, electron-donating groups on P will

*increase σ-donating ability

*decrease π-accepting ability

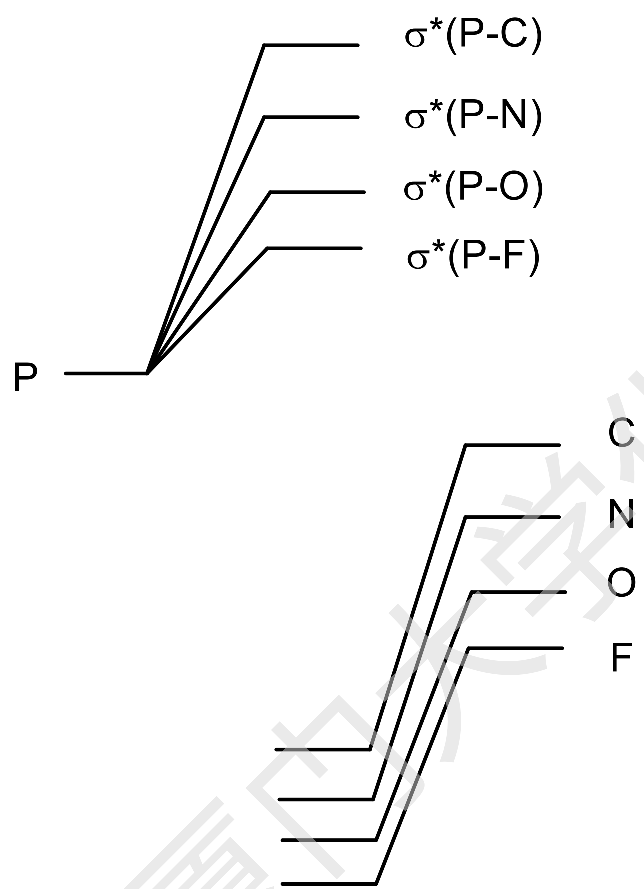
*overall, increase the basicity of PR_3

π acceptor ability

$PR_3 \approx P(NR_2)_3 < PAr_3 < P(OR)_3 < P(OAr)_3 < PCl_3 < CO \approx PF_3$

2. Electronic property of PR_3

Q2. Why is PF_3 a stronger π-acceptor than PMe_3?



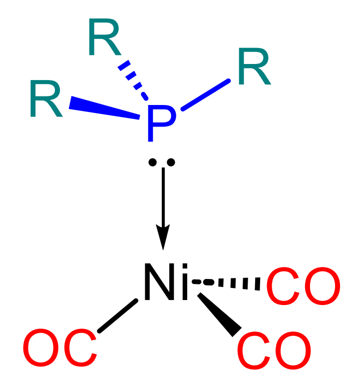
The lower the energy of σ* (P-R), the higher the π–accepting ability of PR_3.

Electron-withdrawing group X make the σ*(P-R) have lower energy, thus increase the π-accepting ability.

The electronic parameter (υ) of phosphine

The electron donating or withdrawing ability of phosphines can be correlated with the CO stretching frequency of monophosphine metal carbonyls. The CO stretching frequency of $LNi(CO)_3$ is called Tolman electronic parameter of L.

L in $L\text{-}Ni(CO)_3$:	νCO
PBu^t_3	2056.1
PMe_3	2064.1
PPh_3	2068.9
$P(OMe)_3$	2079.5
$P(OPh)_3$	2085.0
PF_3	2110.8



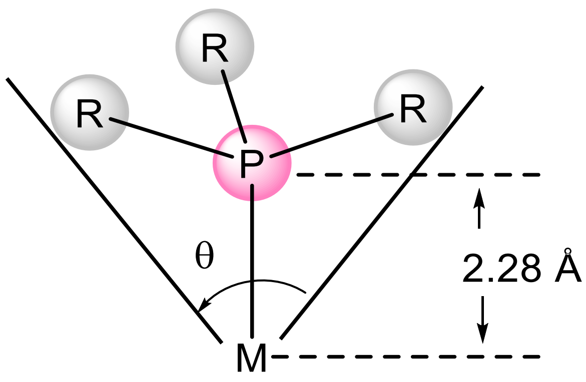
Note different names: PR_3 – phosphine; $P(OR)_3$ - phosphite

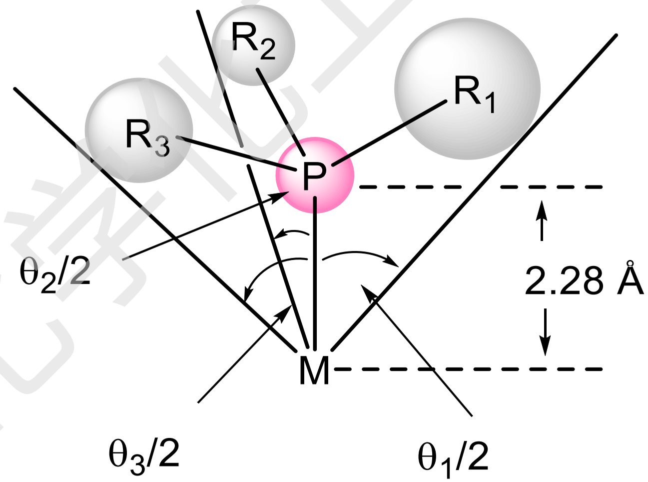
A lower CO stretching frequency mean that the ligand is more basic, more σ -donating, and less π-accepting.

3. Steric property of phosphine ligands

PR_3 Ligands have varying size, depending on R.

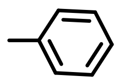
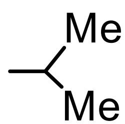
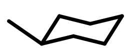
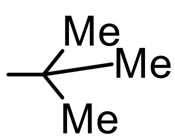
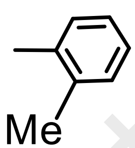
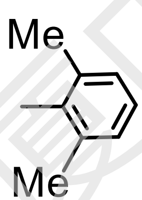
Tolman introduced the concept of *cone angle* to describe the size of PR_3.

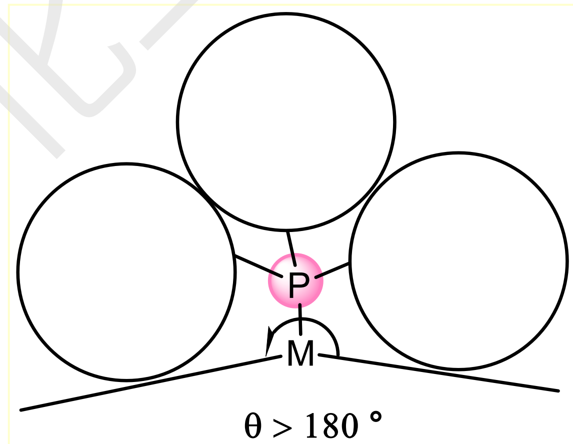




$$\theta = 2/3 \sum_{i=1}^{i=3} \theta_i/2$$

Can a cone angle be larger than 180º ?

PR_3	structure of R	cone angle(θ^o)
$P(OEt)_3$	—O—Et	109
PMe_3	—Me	118
PEt_3	—Et	132
PPh_3		145
$P(i\text{-}Pr)_3$		160
PCy_3		170
$P(i\text{-}Bu)_3$		182
$P(o\text{-}C_6H_4Me)_3$		194
$P(mesityl)_3$		212



Importance of cone angles

- **Cone angle determines the maximum number of ligands that can be attached to a metal. *e.g.***

PCy_3, P(*i*-Pr)$_3$ ----> at most $M(PR_3)_2$ (θ = 170 ^{o}, 160 ^{o})

PPh_3 --------> $M(PPh_3)_3$ or $M(PPh_3)_4$ (rare) (θ = 145 ^{o})

PMe_3 --------> $M(PMe_3)_6$ is possible. (θ = 118 ^{o})

- **Useful when steric effect is considered, *e.g.* in substitution reactions.**

Phosphines tend to be sterically demanding, and thus steric effects dominate coordination

- Bulky phosphines will tend to bind trans to one another

- The presence of several bulky phosphines will result in deviations from ideal geometries.

Summary

Electronic and steric properties of typical phosphines.

- Bulkiness:

$$PBu^t_3 > PCy_3 > PPr^i_3 > PPh_3 > PF_3 > PH_3$$

- Donor Properties:

$$\boxed{PBu^t_3 > PCy_3 > PPr^i_3 > PMe_3} > PPh_3 > \boxed{P(OMe)_3 > PF_3}$$

Good σ-donors,
Poor π–acceptors

Poor σ-donors,
Good π–acceptors

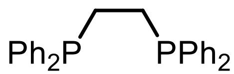
Chelating phosphines

Chelating phosphines are widely used to control structure and reactivity.

Natural Ligand Bite Angle (βn): The ligand preferred P-M-P angle for a chelating diphosphine. This can be determined from x-ray structures or by molecular modeling.

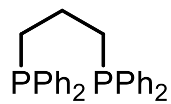
dmpe

72 °



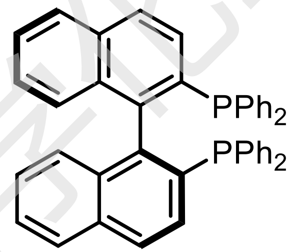
diphos
dppe

85 °



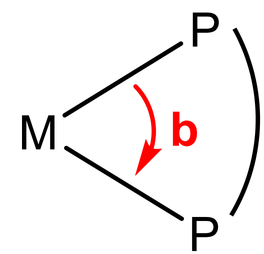
dppp

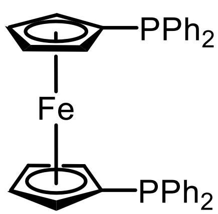
91 °



BINAP

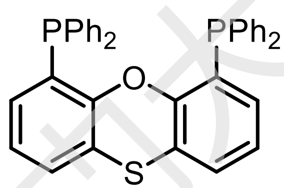
92 °





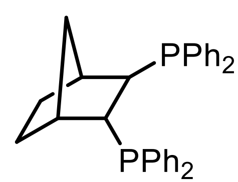
dppf

96 °



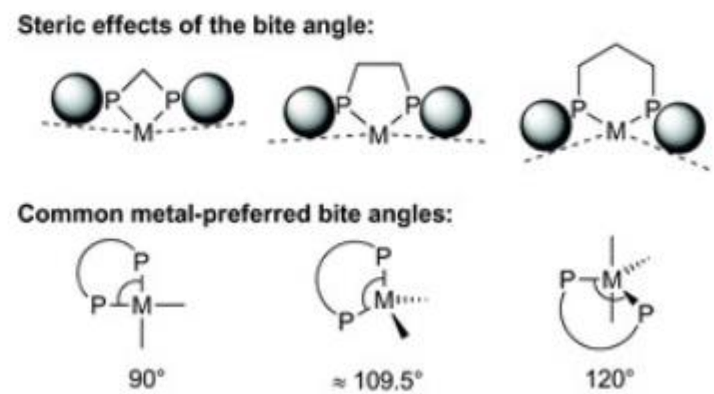
Thisantphos

110 °



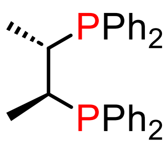
NORPHOS

123 °

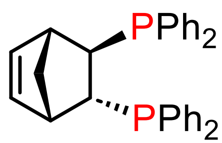


Chiral phosphines

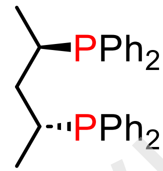
A variety of chiral phosphines are known. For example,



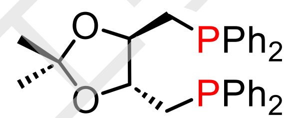
Chiraphos



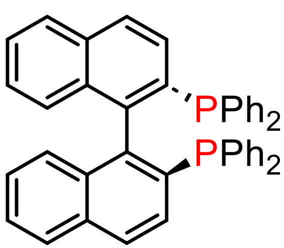
NORPHOS



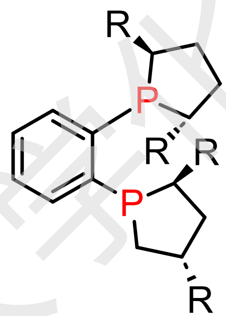
SKEWPHOS



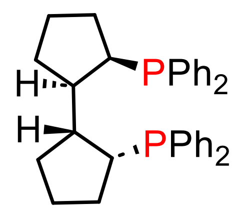
DIOP



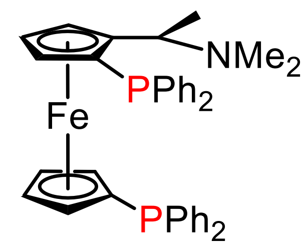
BINAP



DuPHOS

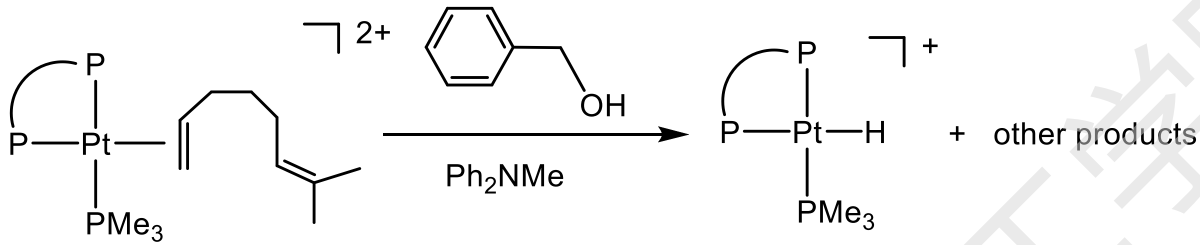


BICP



JOSIPHOS

Practice Problems



Write down the Mechanism and give the other products

HOMEWORK

1. Which of the following complexes would you expect to have the lowest C-O stretching frequency?

(a) $Ni(CO)_4$ (b) $Ni(CO)_3(PPh_3)$

(c) $Ni(CO)_3(PMe_3)$ (d) $Ni(CO)_3(PPhMe_2)$

More e- donationing PR_3 => More electron-rich metal center ==> more backdonation to CO ==> lower ν(C-O)

2. Select the compound with highest ν(CO).

(a) CO (b) F_3B-CO (c) $Fe(CO)_5$

3. Select the compound with the shortest C-O distance.

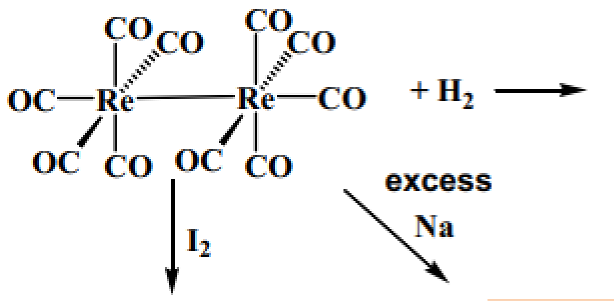
(a) $Mn(CO)_6^+$ (b) $W(CO)_6$ (c) CO

- σ-Donation strengths the C-O bond, decreases C-O distance, increases ν(CO).
- π-Backdonation has the opposite effect.

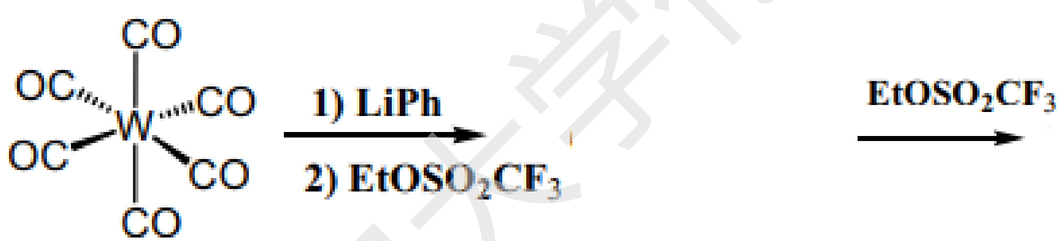
HOMEWORK

4. Provide the products for the following reactions.

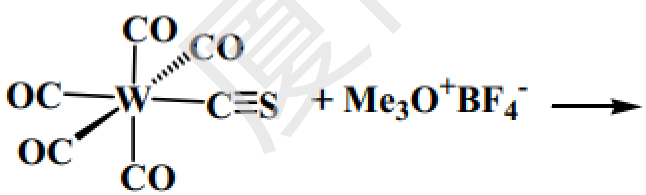
(1)



(2)



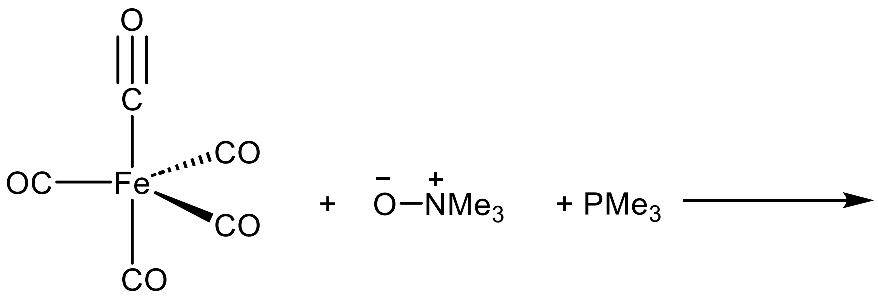
(3)



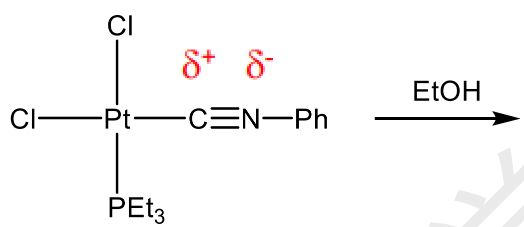
HOMEWORK

5. Provide the products for the following reactions.

(1)



(2)



(3)